CONFIDENCE-AWARE PERSONALIZED ADAPTIVE AUDIO CONTROL

INDIVIDUAL THESIS

SAMARASEKARA K. S. M.

IT22172396

Individual component: Adaptive Audio Controller

Group ID: R26-DS-006

Report submitted in partial fulfilment of the requirements for the

B.Sc. (Hons) in Information Technology

Specializing in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

I declare that this thesis does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis while retaining the right to use its content in future work.

The use of generative AI and automated tools is disclosed in Appendix E. I remain responsible for verifying the accuracy, originality, references, implementation descriptions, data analysis and conclusions in accordance with the applicable university policy.

Student name: Kadiramullage Sarindu Manusara Samarasekara

Student ID: IT22172396

Signature: _____ Date: _____

Supervisor certification. The candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: _____		
Co-supervisor: _____		

ABSTRACT

Virtual reality meditation requires audio that respects individual preferences without introducing distracting or poorly justified changes. This research develops a confidence-aware personalized adaptive audio controller for the Japanese Temple Pond Garden environment. A scene-aware preference pipeline prepares generated instrumental meditation music before playback, while fixed environmental ambience remains separately controlled. A participant-specific disjoint LinUCB contextual bandit selects among 13 bounded actions using a versioned 40-element context. Independent safety filtering limits actions, and delayed learning requires reliable physiological windows collected after actual playback convergence and settling. Three whole-controller conditions support comparison: Static Personalized, Rule-Based Adaptive and Personalized LinUCB. The implementation includes confidence and signal-quality gating, optional HRV handling, participant-isolated persistence, transition logging and participant-level descriptive analysis.

Technical verification passed 22 standalone C# tests and five Python evaluation tests; audio runtime and editor sources also compiled against the checked Unity APIs. A supplementary preference-only simulation compared rule-based, linear bandit, PPO, TD3 and SAC methods across three seeds. The rule baseline achieved the highest simulated composite reward, while SAC achieved the highest preference score, illustrating the importance of multiple outcomes and strong baselines. These simulation results are not evidence for the deployed online agent or for human relaxation. A counterbalanced participant study is specified but has not yet produced results. The individual contribution is an auditable online audio-control architecture that separates generated content, safe playback adaptation and valid delayed evidence. The work establishes technical feasibility within the tested scope and a reproducible basis for investigating personalization, comfort and relaxation-related outcomes.

Keywords: adaptive audio, contextual bandit, virtual reality meditation, physiological confidence, generative music

ACKNOWLEDGEMENTS

I sincerely thank my supervisor, **Mr. Samadhi Rathnayake**, and my co-supervisor, **Ms. Malithi Nawarathne**, for their academic guidance and support throughout this research. I also acknowledge the institutional support provided during the project and the collaboration of research group **R26-DS-006** in developing the shared wearable device and interfaces on which the adaptive audio component depends. The individual work presented in this thesis remains my responsibility.

TABLE OF CONTENTS

1 INTRODUCTION 1
1.1 Overall project context 1
1.2 Individual component context 1
1.3 Problem statement 2
1.4 Individual research gap 2
1.5 Aim and objectives 3
1.6 Research questions and hypotheses 4
1.7 Scope limitations and assumptions 4
1.8 Individual research and product contribution 5
1.9 Thesis structure 5
2 LITERATURE REVIEW AND RELATED WORK 6
2.1 Review method and scope 6
2.2 VR meditation and sound as an intervention context 6
2.3 Physiological measurement and reliability 7
2.4 Contextual bandits for personalization 7
2.5 Deep reinforcement learning alternatives 8
2.6 Safe learning and independent control constraints 9
2.7 Generative audio and playback control 10
2.8 Gap synthesis and conceptual framework 10
3 RESEARCH METHODOLOGY 11
3.1 Research design 11
3.2 Data and study materials 11
3.3 Comparative controller conditions 12
3.4 Prospective participant protocol 12

3.5 Control of learning and carryover 13
3.6 Measures and evaluation metrics 14
3.7 Data analysis 15
3.8 Ethics privacy security and responsible AI 16
3.9 Reproducibility and research artefacts 16
4 PRODUCT AND SOLUTION DESIGN 18
4.1 Functional and nonfunctional requirements 18
4.2 Component architecture 19
4.3 Design alternatives and decisions 20
4.4 Preference profile and scene congruence 21
4.5 State and context specification 21
4.6 Bounded action catalogue 22
4.7 Online LinUCB formulation 23
4.8 Confidence-aware safety design 24
4.9 Delayed outcome attribution 25
4.10 Reward specification 26
4.11 Audio rendering and generated-content management 27
4.12 Persistence and telemetry 28
5 IMPLEMENTATION INTEGRATION AND TESTING 29
5.1 Development environment and technology stack 29
5.2 Profile preparation and startup sequence 30
5.3 Numerical model and policy implementation 30
5.4 Agent lifecycle and outcome state machine 31
5.5 Physiological integration 32
5.6 Playback application and diagnostics 32
5.7 Persistence replay and evidence protection 33

5.8 Verification strategy 33
5.9 Deployment and acceptance procedure 34
6 RESULTS AND VALIDATION 36
6.1 Completed online-agent verification 36
6.2 Numerical and integrity results 37
6.3 Delayed learning and analysis results 37
6.4 Supplementary preference-only simulation protocol 38
6.5 Supplementary algorithm results 38
6.6 Participant and integration validation status 40
7 DISCUSSION 42
7.1 Interpretation of technical findings 42
7.2 Interpretation of algorithm choice 42
7.3 Comparison with literature and baselines 43
7.4 Research contribution 43
7.5 Product contribution 43
7.6 Limitations and threats to validity 44
7.7 Implications for the forthcoming study 45
8 COMMERCIALISATION AND SUSTAINABILITY 46
8.1 Value proposition and target users 46
8.2 Market and competitive position 46
8.3 Operating and sustainability model 47
8.4 Scalability intellectual property and compliance 47
8.5 Contribution to sustainable development 47
9 CONCLUSION AND FUTURE DIRECTIONS 48
9.1 Conclusion 48
9.2 Achievement of objectives 48

9.3 Individual research and product contributions 49
9.4 Immediate and longer-term research 49
REFERENCES 51
GLOSSARY 54
APPENDICES 55
Appendix A Reproducible controller specification 55
Appendix B Physiological and generation contracts 56
Appendix C Participant instruments and operator record 57
Appendix D Reproduction and analysis procedure 58
Appendix E Academic integrity and AI use disclosure 59

LIST OF FIGURES

Figure 3.1 Prospective participant evaluation workflow 13

Figure 4.1 Audio component architecture and evidence boundaries 20

Figure 4.2 Playback-aware delayed physiological attribution 26

Figure 5.1 Decision and delayed-learning state machine..... 31

Figure 6.1 Supplementary synthetic episode reward across three runs..... 39

LIST OF TABLES

Table 2.1 Learning alternatives and their relevance to the audio study 9

Table 3.1 Outcome definitions and units of analysis 14

Table 4.1 Requirements and verification routes 18

Table 4.2 Versioned 40-element bandit context 21

Table 4.3 Audio action catalogue 22

Table 4.4 Principal safety settings 24

Table 5.1 Technology responsibilities 29

Table 5.2 Verification groups and evidence 34

Table 6.1 Completed technical verification results 36

Table 6.2 Supplementary simulation reward and preference results 39

Table 6.3 Supplementary simulation control metrics 40

Table 9.1 Objective achievement and remaining evidence 48

Table A.1 Timing and learning configuration 55

Table B.1 Input fields and audio usage 56

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DSP	Digital Signal Processing
HR	Heart Rate
HRV	Heart Rate Variability
JSONL	JSON Lines
LinUCB	Linear Upper Confidence Bound
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
RMSSD	Root Mean Square of Successive Differences
SAC	Soft Actor-Critic
SD	Standard Deviation
SDNN	Standard Deviation of Normal-to-Normal Intervals
TD3	Twin Delayed Deep Deterministic Policy Gradient
UTC	Coordinated Universal Time
VR	Virtual Reality

1 INTRODUCTION

This thesis presents the design, implementation and technical evaluation of a confidence-aware personalized adaptive audio controller for virtual reality meditation. The system generates a preference-aligned meditation layer and learns how to make bounded playback adjustments using delayed physiological feedback. The completed software evidence and the prospective participant evaluation are reported separately.

1.1 Overall project context

Laminar VR is an immersive meditation research platform that connects a mobile application, wearable sensing, stress inference, a Unity virtual environment and adaptive audiovisual output. Research group R26-DS-006 develops these capabilities as cooperating components. The audio component consumes agreed session and physiological contracts rather than implementing a second wearable acquisition or stress-classification system.

The supported research scene is the Japanese Temple Pond Garden. Its environmental ambient recording remains associated with the scene. Personalized instrumental meditation music is generated separately from a prompt that combines the selected environment and the participant's sound preferences. Keeping these layers separate allows scene consistency and individual musical preference to coexist.

The intended application is supervised wellness research. The controller is not a diagnostic device and does not prescribe treatment. Its engineering objective is to deliver stable, comfortable and personalized sound while preserving sufficient evidence to investigate whether adaptation supports relaxation.

1.2 Individual component context

Component E owns audio preference mapping, scene-aware prompt construction, generated-clip management, adaptive parameter control, local policy learning, safety filtering and audio telemetry. Google Lyria supplies musical content. A Unity

controller decides how that content and the fixed ambience are played. These responsibilities are deliberately separate: the learning agent selects control actions, not notes, lyrics or waveforms.

The learned controller is a participant-specific disjoint Linear Upper Confidence Bound contextual bandit, abbreviated LinUCB. It selects one safe micro-adjustment from a catalogue and updates its outcome estimate after an eligible physiological observation. The controller operates online inside Unity without requiring a neural-network training service during the session.

1.3 Problem statement

Fixed meditation recordings provide the same behaviour even when users prefer different instruments, pacing, brightness or music-to-ambience balance. A fixed recording can be personally appropriate at session start yet become too prominent, repetitive or distracting as the participant settles. Preference-aware initialization alone does not address this changing context.

Adaptation also introduces risk. Stress estimates may be uncertain, physiological measurements may be degraded, and feedback arrives after an aggregation window rather than immediately. An agent that treats every frame as a new physiological outcome can learn from repeated or overlapping evidence. An agent that responds strongly to uncertain input can cause unnecessary or uncomfortable changes.

The research problem is therefore to learn personalized audio adjustments from sparse, delayed and reliability-qualified feedback while protecting playback continuity. The solution must distinguish a proposed action from the action actually executed and must not attribute an outcome to a change that was never heard.

1.4 Individual research gap

Published VR mindfulness research provides evidence about feasibility, adherence and intervention outcomes [1], [2]. Experimental music research shows that stress-related responses are not uniform across physiological and subjective measures [3]. These findings motivate adaptive sound but do not establish a validated policy for confidence-aware, participant-specific audio control.

Contextual bandits supply a practical method for learning personalized action preferences from partial feedback [5]. Safe learning research explains why unconstrained reward maximization is insufficient when actions affect people [6]. The implementation gap addressed here is their combination with scene-aware generated meditation content, fixed ambience, delayed physiological windows, actual-playback verification and auditable local learning.

The contribution is an integrated audio-control architecture and an explicit evaluation protocol. It is not a claim that confidence gating, contextual bandits or generative music are individually new. The research questions concern whether their combination is useful and measurable in this particular meditation setting.

1.5 Aim and objectives

The aim is to develop and technically evaluate a confidence-aware personalized audio controller that can learn conservative playback adaptations during VR meditation and support a controlled investigation of relaxation-related outcomes.

- Represent persistent sound preferences as a normalized audio profile and personalized baseline.
- Generate instrumental meditation music consistent with the selected environment before session playback begins.
- Construct a finite, versioned context from preferences, applied audio state, physiological reliability and recent history.
- Implement participant-specific online LinUCB learning with bounded, interpretable actions.
- Enforce deterministic safety, smooth transitions and valid delayed-outcome attribution.
- Compare Static Personalized, Rule-Based Adaptive and Personalized LinUCB controller conditions.
- Verify numerical behaviour, failure handling, persistence and evaluation aggregation using automated tests.
- Specify an ethical, reproducible participant study without substituting simulation results for human evidence.

1.6 Research questions and hypotheses

RQ1 asks whether a scene-aware preference pipeline can initialize an appropriate meditation and ambience configuration. RQ2 asks whether a local contextual bandit can learn useful participant-specific control choices from valid delayed feedback. RQ3 asks whether confidence and signal-quality gating can prevent learning from unreliable or contaminated observations.

RQ4 compares the three whole-controller conditions in relaxation-related change, perceived personalization, smoothness and intervention burden. RQ5 examines the operational cost of conservative attribution, including how often an action receives no usable outcome.

The prospective primary hypothesis is that Personalized LinUCB will produce a greater participant-level reduction in reported session stress than Static Personalized. A second comparison examines LinUCB against Rule-Based Adaptive. These are hypotheses to test, not established findings. Comfort and distraction are separate outcomes because an improved controller reward need not imply a better participant experience.

1.7 Scope limitations and assumptions

The implementation covers one scene, instrumental generated meditation audio, fixed environmental ambience and seven normalized playback-control values. Because music mix and ambience mix are complementary, these values contain six independent control dimensions. The contextual bandit has 13 actions and a 40-element context.

Physiological input is supplied by Component B, normally at 60-second window boundaries. The controller assumes correct participant routing, coherent timestamps and an externally valid signal-quality estimate. Missing optional heart-rate variability values are not reconstructed. Real-user data collection and headset deployment validation remain outside the completed technical results reported in this submission.

The music-generation path uses asynchronous clip requests, caching and crossfading. Playback control is real time; uninterrupted provider-side streaming music generation

is not assumed. The system does not infer that short-window heart-rate variability alone proves physical relaxation.

1.8 Individual research and product contribution

The individual contribution comprises the audio profile and prompt pipeline, the bounded control catalogue, the online LinUCB implementation, the confidence-aware safety layer, the delayed-outcome gate, the reward specification and participant-isolated persistence. It also includes Unity audio integration, generated-content startup control, decision logging and participant-level analysis tools.

The sensing hardware, stress-prediction internals and mobile account service are integration dependencies, not claimed audio inventions. The thesis concentrates on how their accepted outputs are interpreted and used by the audio component. The research contribution is supported by explicit tests and traceable implementation decisions rather than a claim of demonstrated clinical effectiveness.

1.9 Thesis structure

Chapter 2 reviews the relevant evidence and learning methods. Chapter 3 defines the research methodology and participant protocol. Chapter 4 explains the architecture, mathematical model and design choices. Chapter 5 documents implementation and verification. Chapter 6 reports available results. Chapters 7 to 9 discuss interpretation, sustainability, conclusions and further research. The appendices provide reproducibility specifications and study instruments.

2 LITERATURE REVIEW AND RELATED WORK

The review connects VR meditation, music-related stress responses, physiological measurement and safe personalized decision making. It distinguishes evidence motivating the design from evidence that would be needed to validate the implemented controller.

2.1 Review method and scope

The review is a focused technical narrative rather than a systematic review or meta-analysis. Primary publications and official technical documentation were selected for direct relevance to the implemented method. Sources cover VR mindfulness studies, music and stress experiments, HRV research guidance, contextual bandits, safe learning and continuous-control algorithms.

The review does not claim an exhaustive database search, a registered review protocol or a screening count. The central selection criterion was whether a source explained a design assumption, supported a measurement decision or provided a credible alternative controller. Algorithm suitability is interpreted in the context of sparse human feedback, not transferred uncritically from robotics or recommendation benchmarks.

2.2 VR meditation and sound as an intervention context

Modrego-Alarcon and colleagues investigated mindfulness programmes with and without VR support in a randomized university-student study [1]. Their results support mindfulness-related stress improvements and suggest that VR can support adherence. They do not isolate adaptive audio as the mechanism. Consequently, evidence for VR-supported mindfulness cannot be reported as evidence for this controller.

Navarro-Haro and colleagues investigated the feasibility and acceptability of VR mindfulness with meditation-conference participants [2]. That study supports the plausibility of the immersive setting, but its participants and intervention differ from

the present system. It motivates measuring acceptance and comfort alongside technical output rather than assuming that immersion guarantees benefit.

Thoma and colleagues experimentally compared relaxing music, rippling-water sound and rest before a stress task [3]. The effects differed across measured response domains. This is important for the present design: a change in one stress-related proxy should not be presented as a universal improvement in relaxation. Physiological and self-reported outcomes are therefore retained separately.

2.3 Physiological measurement and reliability

Heart rate is measured in beats per minute. RMSSD summarizes successive beat-interval differences, while SDNN summarizes the dispersion of normal-to-normal intervals. Laborde, Mosley and Thayer emphasize recording context, baseline, artefacts and transparent reporting when interpreting HRV [4]. These considerations support careful window qualification and consistent measurement duration.

The audio controller distinguishes predictive confidence from signal quality. Predictive confidence describes the certainty attached to the stress estimate. Signal quality describes the usability of the underlying measurement window. Neither is a physiological improvement measure. A high-confidence prediction can still be wrong, and a high-quality signal can still produce an uncertain stress estimate.

The controller uses these values to limit action eligibility and learning. This is an engineering response to uncertainty, not proof of calibrated probabilities or measurement accuracy. Confidence calibration, short-window HRV validity and sensor artefact rejection require separate empirical examination.

2.4 Contextual bandits for personalization

Li and colleagues describe contextual action selection with partial feedback and present LinUCB for personalized recommendation [5]. For each decision, the learner observes context, chooses an action and receives feedback only for that choice. The method balances estimated reward with uncertainty. This structure is useful when a system cannot observe what would have happened under every alternative action.

In adaptive audio, the context includes preference and physiological information, and the action is a bounded adjustment. The feedback arrives after playback and measurement delays. The bandit formulation estimates the expected outcome of each action in the current context; it does not model an entire meditation session as a long-horizon transition process.

This distinction limits the claim. Physiological carryover and slowly changing user state may violate simple bandit assumptions. The method is selected for interpretability and manageable online updates, not because it is guaranteed to recover the causal effect of music or to outperform all deep learning approaches.

2.5 Deep reinforcement learning alternatives

Proximal Policy Optimization uses policy-gradient updates with a clipped surrogate objective to restrict large policy changes [8]. It is a credible continuous-control comparison when sufficient representative interaction data are available. However, conservative policy optimization does not automatically enforce safe audio changes, and on-policy updates can be demanding when human outcomes arrive slowly.

Twin Delayed Deep Deterministic Policy Gradient reduces actor-critic estimation problems using twin critics and delayed policy updates [9]. Its off-policy design allows data reuse. For this research setting, the practical burden includes replay design, exploration control, reward stationarity and safe handling of continuous actions with limited participant data.

Soft Actor-Critic combines off-policy learning with an entropy-based objective [10]. It is an important candidate for richer continuous-control research. Its stochastic exploration still needs an independent safety layer, and its suitability for sparse real-user audio feedback must be measured. The architecture does not assume that SAC will outperform PPO, TD3 or a well-designed rule controller.

Table 2.1 Learning alternatives and their relevance to the audio study

Method	Learning characteristic	Implication for this study
Static Personalized	No adaptation or learning	Controls for preference-aligned content alone
Rule-Based Adaptive	Fixed stress-aware decisions	Explainable non-learning adaptive comparator
LinUCB	Online linear reward estimates per action	Local interpretable updates from qualified outcomes
PPO	On-policy neural policy optimization	Suitable simulation comparison; greater interaction demand
TD3	Off-policy deterministic actor with twin critics	Continuous-control candidate requiring careful exploration
SAC	Off-policy stochastic maximum-entropy actor	Flexible candidate; entropy is not a safety mechanism

2.6 Safe learning and independent control constraints

Garcia and Fernandez review approaches that incorporate safety into reinforcement learning [6]. The relevant design principle is that a reward function alone is insufficient protection: undesirable actions can still be explored before penalties are learned. The audio component therefore places a deterministic filter outside the learned model.

The filter defines the permitted action set before selection and checks the selected action again before application. This separates model uncertainty from operational permission. A highly rewarded or highly uncertain action is not eligible if it violates control limits. Holding the current safe state is a legitimate action rather than a failure to demonstrate adaptation.

2.7 Generative audio and playback control

Google documents Lyria clip generation and a separate real-time music interface [13]. The implemented clip service supplies generated instrumental content from a scene-aware prompt. Its availability and latency are external operating constraints. The learning policy does not control the generator's internal model or guarantee its musical output.

Unity provides playback controls such as AudioSource pitch [14]. The prototype maps normalized audio controls onto gain, filtering, pitch, smoothing and crossfading. These mappings are practical approximations: pitch adjustment is not independent beat-preserving tempo modification, and a density control over a mixed clip does not create or remove individual instrumental stems.

2.8 Gap synthesis and conceptual framework

The review motivates a system in which preference alignment establishes the starting configuration, physiology qualifies when adaptation is meaningful, and bounded online learning selects small adjustments. A separate safety layer and delayed-outcome gate protect the transition from decision to evidence.

The conceptual framework links four constructs: personal preference, reliable current context, conservative intervention and measurable delayed outcome. The research gap is whether this combination provides a useful adaptive audio mechanism in a VR meditation session. That question requires controlled participant evidence beyond software correctness and simulated reward.

3 RESEARCH METHODOLOGY

The research combines design-science development, deterministic verification, supplementary simulation and a prospective controlled participant study. Each evidence stream answers a different question and is analysed at its appropriate unit of observation.

3.1 Research design

The development follows a design-science orientation: identify a practical problem, define objectives, construct an artefact and evaluate its behaviour [7]. The artefact includes a decision model, a runtime integration and an evidence pipeline. The evaluation separates algorithmic correctness from user experience and relaxation-related outcomes.

Technical verification addresses whether the implementation follows its specification. Supplementary simulations investigate control trade-offs under an explicit synthetic preference model. The planned participant study addresses acceptability and comparative session outcomes. No human efficacy conclusion is inferred from either software test success or synthetic reward.

3.2 Data and study materials

Persistent audio preferences include preferred instruments, mood, tempo, intensity, music-to-ambience balance, brightness, novelty tolerance, dissonance avoidance, rhythm, nature content, spaciousness and volume. The selected scene supplies the environmental context. The production audio preference manager loads a configured user profile; participant routing and stable pseudonymous identities must be verified before recruitment.

The physiological contract contains a timestamp, a stress estimate and confidence, signal quality, window start and end, and optional HR, RMSSD and SDNN. The audio adapter consumes accepted windows from the integration bridge. The nominal update is one window per 60 seconds, not one new physiological observation per rendering frame.

The study materials are the Japanese Temple Pond Garden scene, its fixed ambience, a generated meditation clip, the same playback mapping and the same safety configuration across controller conditions. Headset, headphones, operator instructions and listening settings must be held consistent and documented.

Long-term preference and temporary session state have distinct meanings. Baseline stress, fatigue, sleep and comfort can be collected for the study without overwriting the participant's usual audio preferences. The active bandit context contains its implemented 40 features; these questionnaire items must not be described as model inputs unless an explicit implemented adapter supplies them.

3.3 Comparative controller conditions

Static Personalized initializes from the preference pipeline and selects NoChange at scheduled decision opportunities. Rule-Based Adaptive uses a fixed conservative stress-aware controller, discretized to the shared audio action catalogue. Personalized LinUCB selects an eligible catalogue action using its participant-specific reward estimates and uncertainty bonus.

All three conditions use the same content, normalized controls, playback mapping, observation qualification and outcome logging. Only the learned condition updates model weights. The rule controller is an independent comparator, not a correction added to LinUCB. A fixed condition is assigned for each research session.

3.4 Prospective participant protocol

The proposed evaluation is a supervised within-participant pilot with counterbalanced condition order. Six permutations of the three conditions distribute order effects where recruitment permits. The final sample size, recruitment criteria and session duration require supervisor and institutional approval. No sample size or ethics approval is asserted as already secured.

Each participant receives an explanation of the system, data handling, generated content and voluntary withdrawal. After consent, the operator assigns a pseudonymous identity, records preferences and checks hearing comfort. A personalized generated clip is prepared before session playback. A failed generation

request does not permit substitution of unrelated audio as a successful personalized session.

The session includes acclimatization, an adaptive exposure period and stabilization. The controller requires at least two reliable independent baseline windows. A proposed exposure duration of 12 to 15 minutes may yield only a small number of qualified updates; the pilot must confirm feasibility before finalizing this duration. Session timing is identical across conditions.

After each session, the participant reports current stress, perceived relaxation, audio personalization, transition smoothness, distraction and discomfort. A recovery break is provided and the next session proceeds only with renewed willingness. The operator records dropouts, interruptions, content failures and adverse experiences, not only completed sessions.

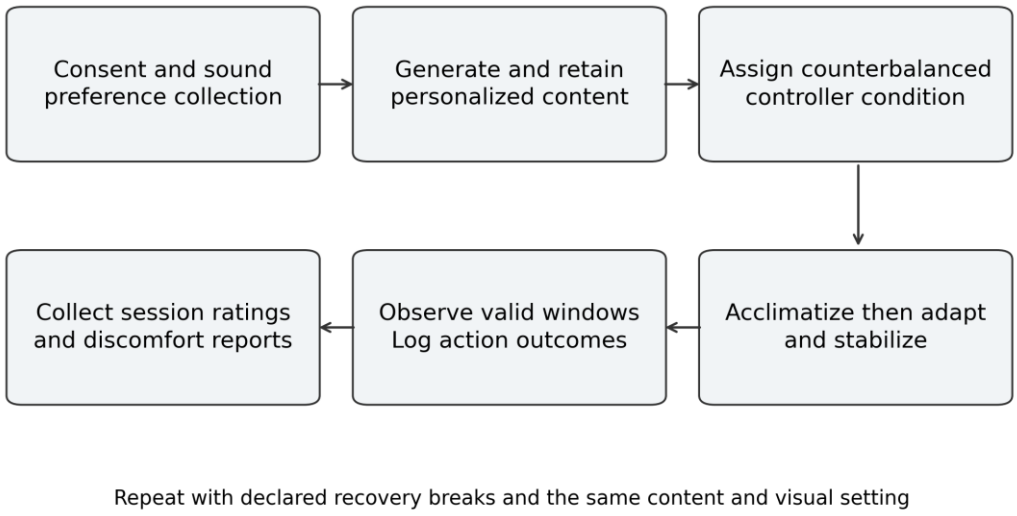


Figure 3.1 Prospective participant evaluation workflow

3.5 Control of learning and carryover

Online-learning feasibility and frozen-policy performance answer different questions. The primary pilot evaluates the online-learning package with its exploration and persistence settings declared in advance. Initial and final snapshots are retained to

describe how much the model changed. A separate frozen evaluation can disable weight updates and exploration to examine the learned greedy policy.

No model trained on simulated observations is loaded into the real-user domain. For first-exposure comparisons, participant models begin from the declared ridge prior. For longitudinal personalization, the same participant's model can persist across scheduled visits. These two protocols must not be mixed without reporting model exposure and session order.

The same generated clip should be reused across a participant's condition comparisons when the purpose is to isolate playback control. Visual adaptation is held fixed for the audio-specific experiment. Composition replacement, unrelated environment changes and manual interventions invalidate affected online outcomes and are separately documented.

3.6 Measures and evaluation metrics

The proposed primary human endpoint is the participant-level pre-to-post change in self-reported current stress on a consistently anchored 0 to 10 scale. Secondary outcomes include perceived relaxation, personalization, smoothness, distraction and discomfort. These research items are not presented as validated clinical instruments.

Physiological analyses report the accepted stress-score trajectory and optional HR and RMSSD changes independently of the composite reward. For HRV, recording duration, missingness, artefact handling and any planned transformation are declared. A 60-second window is not treated as interchangeable with a conventional longer recording [4].

Table 3.1 Outcome definitions and units of analysis

Outcome	Definition	Analysis unit
Reported stress change	Post-session score minus pre-session score	Participant and condition
Perceived audio quality	Anchored personalization, smoothness and distraction items	Participant and condition

Outcome	Definition	Analysis unit
Mean valid reward	Average composite reward over eligible outcomes	Session then participant
Stress improvement	Pre-window stress minus eligible post-window stress	Valid action outcome
Preference alignment	Distance-based proximity to the initialized profile	Session then participant
Safety-filter intervention	Fraction of decisions whose proposal was altered	Session then participant
Control intervention frequency	Fraction of decisions with a nonzero safe action	Session then participant
Invalid-outcome rate	Invalid or cancelled outcomes divided by resolved outcomes	Session then participant
Learning exposure	Valid online updates and eligible actions per session	Participant and condition

3.7 Data analysis

Repeated physiological windows are not independent participants. The implemented analysis utility first summarizes sessions, then averages repeated sessions within participant and condition before reporting between-participant means and sample standard deviations. A standard deviation is missing when fewer than two contributing participants exist.

The prospective inferential analysis uses paired condition contrasts for the primary endpoint. Participant-level paired bootstrap intervals or a repeated-measures model can describe uncertainty after checking the sample and distribution. If both LinUCB-versus-static and LinUCB-versus-rule contrasts are confirmatory, their multiplicity is controlled using a declared procedure such as Holm adjustment. This inferential stage is not implemented by the current descriptive CSV exporter.

Invalid outcomes are excluded from reward means but retained in coverage and invalidity metrics. An invalid-only session does not receive an artificial zero reward. Missing physiology is not imputed as relaxation. Source-domain labels, participant counts and contributing counts for each metric accompany every comparison.

Learning curves plot outcomes against valid update count and elapsed exposure. Session reward alone can favour methods with different numbers of valid windows, so cumulative reward is interpreted alongside coverage. No off-policy estimate is claimed from ordinary logs: statistically valid replay requires appropriate action support and logging assumptions [17].

3.8 Ethics privacy security and responsible AI

Recruitment requires university approval and informed consent. Participant wellbeing, withdrawal rights and transparent risk communication guide the protocol [15]. No approval identifier or completed institutional review is claimed.

The operator provides immediate stop and mute controls and monitors discomfort, dizziness, distress and sensory sensitivity. Numerical bounds do not establish safe sound-pressure exposure; headphone gain and acoustic output require separate checks.

Logs use pseudonymous identifiers, with identity mappings stored separately under access control. API credentials remain on the backend. Prompts exclude unnecessary physiological measurements. Transport security, authentication and retention require verification on the deployed installation.

Participants are told that audio is generated. Prompts request original instrumental content, not artist imitation. Content, filtering, billing and service failures remain operational risks. The system is a research wellness prototype, not a validated clinical intervention.

3.9 Reproducibility and research artefacts

Reproducibility depends on the code revision, feature and action schemas, reward weights, timing settings, content hashes, participant routing and learning snapshots. A generated clip is retained because a prompt alone does not guarantee identical

music on a later request. Simulation seeds and configuration files are stored separately from participant evidence.

The repository provides a standalone C# verification script and Python evaluation tests. JSONL decisions and outcomes support inspection of valid updates and cancellations. A bounded replay buffer is an audit structure, not a neural experience-replay training mechanism. Appendix D lists operational commands and the files required for reproducible analysis.

4 PRODUCT AND SOLUTION DESIGN

The architecture separates content generation, state interpretation, decision making, deterministic safety, playback and research evidence. This chapter defines both the mathematical controller and the practical constraints that govern its use.

4.1 Functional and nonfunctional requirements

The component must start from a valid personalized profile, obtain the generated meditation layer before playback and preserve the fixed environment ambience. It must adapt only through documented normalized controls and must expose enough telemetry to explain why a change was applied or withheld.

Table 4.1 Requirements and verification routes

ID	Requirement	Verification route
FR1	Load preferences and selected scene context	Profile and prompt inspection
FR2	Require generated meditation content before playback	Startup and generation-failure check
FR3	Consume qualified physiological windows	Adapter and outcome-gate tests
FR4	Support three whole-controller conditions	Mode and non-learning tests
FR5	Learn participant-specific eligible action outcomes	Numerical and delayed-update tests
FR6	Mix meditation and fixed ambience smoothly	Unity playback acceptance check
FR7	Persist isolated models and complete transition evidence	Snapshot and logging tests
NFR1	Reject invalid actions and uncertain unsafe adaptation	Safety and action-mask tests

ID	Requirement	Verification route
NFR2	Avoid duplicate or contaminated learning	Attribution and cancellation tests
NFR3	Remain explainable and locally executable	Diagnostics and dependency inspection
NFR4	Protect identity and credentials	Deployment security review

4.2 Component architecture

The preference manager and profile engine produce the initial audio configuration. The prompt builder combines that profile with the Japanese Temple context. The local Lyria service requests and stores generated music, and Unity decodes the returned clip. A separate source owns the environment ambience.

The physiological adapter forwards accepted Component B windows into the audio state. The audio agent encodes context, chooses an action, invokes the safety filter and provides target parameters to the mixer. The mixer reports applied parameters back to the agent. The delayed-outcome gate uses that feedback to decide when a subsequent physiological window can be attributed.

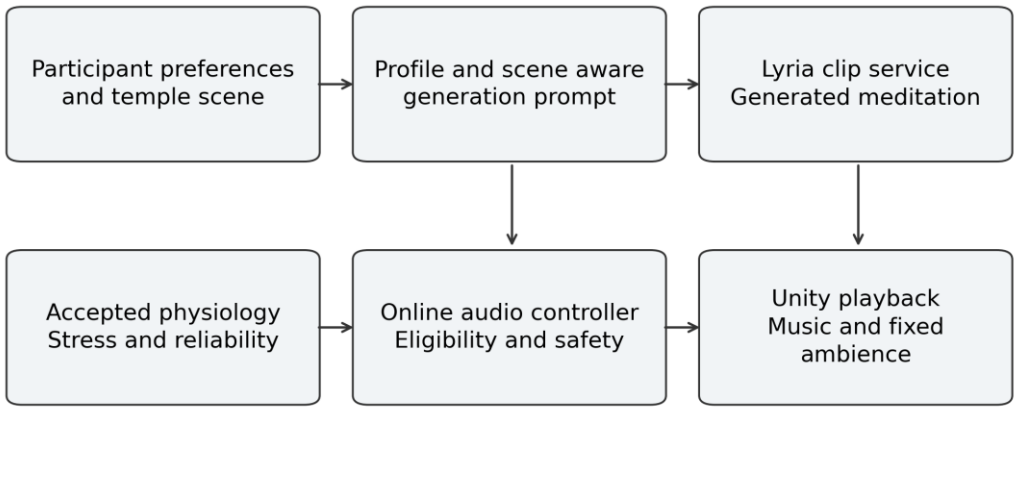


Figure 4.1 Audio component architecture and evidence boundaries

The transition logger records decisions immediately and outcomes when they resolve. Persistence stores only the participant's learning state with its schema and configuration identity. The dashboard observes these stages; it does not bypass the safety path or manufacture a reward when no valid outcome exists.

4.3 Design alternatives and decisions

LinUCB is selected because the action set is small, outcomes are sparse, numerical updates are inspectable and the model can run locally. A neural actor is not required to select among 13 defined alternatives. The model makes its expected reward and uncertainty available for explanation and logging.

This is a contextual-bandit choice, not a statement that deep reinforcement learning is unsuitable in general. PPO, TD3 and SAC remain credible simulation comparators [8]-[10]. Their richer policies are useful when representative sequential data and a justified reward are available. The deployment decision prioritizes auditability and constrained online adaptation under the present data rate.

The model selects a whole action. The rule controller is a comparison condition rather than an action added to the learned proposal. This makes the experimental

contrast clearer: differences arise from controller decisions under the same action and safety interface, not from a hidden combination of two active policies.

4.4 Preference profile and scene congruence

The profile engine maps categorical selections and normalized preference values to a starting parameter vector. Examples include slow, medium and fast tempo targets, darker or brighter sound, and mostly ambience versus mostly music. Profile normalization and conservative range limits are applied before playback.

The current preference encoding has 13 values: baseline intensity, density, brightness, tempo, music mix, ambience mix, rhythm, nature level, reverb, novelty tolerance, dissonance allowance, a responsiveness prior and confidence sensitivity. The last two are profile-derived priors, not measured participant traits. Their names must not imply that physiological responsiveness has been clinically estimated.

The scene context constrains content selection. A preference for piano or flute can shape the meditation layer without changing the identity of the temple environment. The fixed ambience retains pond and garden context. This avoids asking participants to choose a contradictory environmental sound style separately from the selected VR scene.

4.5 State and context specification

The structured state contains the profile, personalized baseline, current applied controls, signal packet, trends and recent safe action. The bandit encoder constructs a fixed 40-element vector and rejects non-finite values. Its finite features are clipped to the interval from -1 to 1, with unit-interval values used where appropriate.

Table 4.2 Versioned 40-element bandit context

Indices	Feature group	Encoding
0	Bias	Constant 1
1 to 13	Preference profile	Thirteen normalized profile values
14 to 20	Applied audio controls	Seven current control values

Indices	Feature group	Encoding
21 to 27	Baseline deviation	Current minus personalized baseline
28 to 30	Stress and reliability	Stress, confidence and signal quality
31 to 32	Recent trends	Stress and confidence differences
33 to 34	Optional physiology	HR divided by 200 and RMSSD divided by 200
35 to 36	Availability	HR and RMSSD presence flags
37	Session progress	Normalized decision progress
38	Novelty exposure	Change count divided by 20
39	Recent action	Maximum safe magnitude divided by action limit

The progress feature follows decision count divided by the configured horizon, rather than an independently validated physiological session clock. SDNN is retained in the signal packet but is not an active bandit feature or reward component. Explicit availability flags prevent a missing HRV value from being interpreted as a genuine zero measurement.

4.6 Bounded action catalogue

The action catalogue contains hold and paired increases and decreases for six independent dimensions. Each ordinary micro-adjustment changes one independent control. A mix action changes music and ambience in opposite directions, preserving their complementary relationship. The default step is 0.03 in normalized units.

Table 4.3 Audio action catalogue

Arm IDs	Action family	Nonzero control changes
0	Hold stable	None
1 and 2	Intensity	Plus or minus 0.03 intensity

Arm IDs	Action family	Nonzero control changes
3 and 4	Density	Plus or minus 0.03 density
5 and 6	Brightness	Plus or minus 0.03 brightness
7 and 8	Tempo	Plus or minus 0.03 tempo
9 and 10	Fade	Plus or minus 0.03 fade
11 and 12	Music and ambience balance	Opposing 0.03 changes to both mix values

An action has seven logged output values even though the two mix values are coupled. The absolute filter limit is 0.08 per value, not the normal exploration step. Hold is arm zero and wins a numerical score tie, providing a conservative deterministic cold-start choice.

4.7 Online LinUCB formulation

For each eligible arm a, the model stores a 40 by 40 covariance matrix A_a and a 40-element reward vector b_a. Initialization uses a ridge prior with lambda equal to 1 and b_a equal to zero. The model estimates a separate linear response for each action [5].

$$A_a = \lambda I,\ b_a = 0,\ \theta_a = solve(A_a, b_a) \quad (4.1)$$

$$score_a(x) = \theta_a^T x + \alpha \sqrt{x^T\ solve(A_a, x)} \quad (4.2)$$

The default exploration coefficient alpha is 0.25. The first term estimates reward in the current context. The second increases the score of uncertain eligible actions. This uncertainty is model uncertainty, distinct from the physiological confidence used by the safety layer.

$$a_t = argmax\ over\ eligible\ a\ of\ score_a(x_t) \quad (4.3)$$

After a valid delayed reward, only the executed catalogue arm is updated. The context is the frozen pre-action vector, not a newly encoded post-action state. Invalid outcomes produce no numerical observation.

$$A_a \leftarrow A_a + x_t\, x_t^T,\ b_a \leftarrow b_a + r_t\, x_t \quad (4.4)$$

Cholesky decomposition solves the systems without explicitly forming an inverse. The implementation validates feature dimensions, finite values, matrix symmetry and positive definiteness. Frozen evaluation sets the exploration contribution to zero and disables updates. There is no Bellman target, actor network or discounted return in this controller.

4.8 Confidence-aware safety design

Every catalogue candidate is checked against the current state and safety configuration. A candidate that would be modified is masked from selection. Hold remains available. The selected action then passes through the same filter independently before application, protecting against state changes between eligibility and execution.

The filter rejects non-finite actions, clamps magnitude, applies confidence and signal-quality restrictions, constrains change acceleration and limits baseline drift. It also preserves normalized mix values with ambience at least 0.15 and music at most 0.85. These are provisional engineering settings, not experimentally established comfort thresholds.

Table 4.4 Principal safety settings

Setting	Default	Meaning
Micro-action step	0.03	Normal catalogue adjustment
Absolute action limit	0.08	Maximum change per control
Action acceleration limit	0.05	Limit relative to preceding safe change
Output slew	0.08 per second	Target movement constraint
Mean baseline distance	0.30	Permitted average deviation
Confidence freeze and low threshold	0.25 and 0.45	Hold and conservative gating

Setting	Default	Meaning
Quality freeze and low threshold	0.30 and 0.60	Hold and conservative gating
Ambient minimum and music maximum	0.15 and 0.85	Complementary mix bounds

The agent also requires sufficient confidence and quality before creating a learnable decision. Therefore the integrated online agent can hold entirely under low reliability rather than necessarily applying a dampened action. Hold cannot accidentally carry forward an earlier nonzero adjustment through acceleration or recovery logic.

If the filter changes a proposal, the proposed arm is not credited with the replacement's outcome. An altered action that does not match a catalogue arm cannot update the fixed-action bandit. Safety-filter intervention is logged; it is not automatically evidence of an unsafe audible event.

4.9 Delayed outcome attribution

The controller maintains at most one pending decision. A valid pre-window supplies context, the action is applied gradually, and playback feedback confirms that the mixer reached its target. Only then does the settling interval begin. A post-window must start after settling and must not overlap the pre-window.

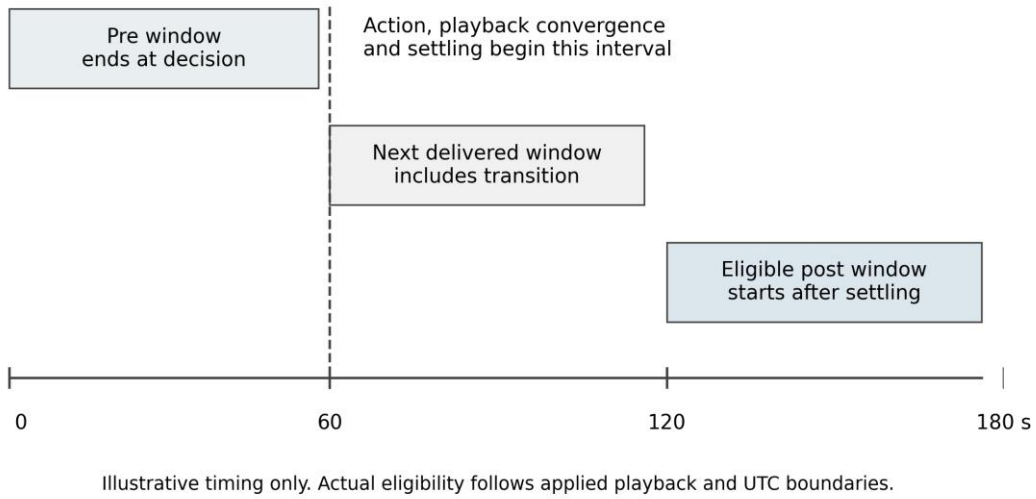


Figure 4.2 Playback-aware delayed physiological attribution

The default production minimum decision interval is 75 seconds, settling is 5 seconds and outcome timeout is 240 seconds. Accepted window durations range from 30 to 120 seconds, with maximum age 90 seconds and future timestamp tolerance 2 seconds. These conditions reject stale, duplicate, overlapping and out-of-domain observations.

With non-overlapping 60-second windows, the first delivery after an action often includes transition time and is ineligible. A later full window may be the first usable outcome. Thus a 75-second minimum does not guarantee an action every 75 seconds, and the learner does not update at headset frame rate.

The same post-window cannot reward two decisions. It may become the pre-window for a subsequent decision. Pause, completion, emergency mute, connection loss, clip replacement or an unrelated environment change cancels the pending attribution. Cancellation records a reason rather than a fabricated physiological penalty.

4.10 Reward specification

The reward combines reliability-weighted physiological change with preference proximity, stability and intervention penalties. Its role is to guide the controller, not to diagnose relaxation. A pre-action and post-action pair is necessary before the reward calculator is invoked for a valid learning observation.

Baseline variation supplies normalization scales with lower bounds of 0.05 for normalized stress, 5 milliseconds for RMSSD and 5 beats per minute for HR. These floors prevent division by a near-zero sample standard deviation. Two baseline windows support initialization but do not provide a robust physiological calibration.

$$\rho = min(c_{pre}, c_{post})\, min(q_{pre}, q_{post}) \quad (4.5)$$

$$z_s = clip((s_{pre} - s_{post}) / \sigma_s, -1, 1) \quad (4.6)$$

$$z_r = clip((RMSSD_{post} - RMSSD_{pre}) / \sigma_r, -1, 1) \quad (4.7)$$

$$z_h = clip((HR_{pre} - HR_{post}) / \sigma_h, -1, 1) \quad (4.8)$$

The HR and RMSSD terms are zero when either required measurement is absent. Preference proximity P equals one minus the mean absolute distance between post-action controls and the personalized baseline. It is a profile-consistency proxy, not a participant satisfaction rating.

Let m be mean absolute safe-action magnitude divided by 0.08 and clipped to the unit interval. Stability S equals 1 minus m. Abruptness A equals m squared. Low-confidence overreaction L equals m multiplied by one minus pre-confidence. Novelty N is the clipped cumulative change count divided by 20. Unnecessary intervention U increases with m when pre-stress is below 0.40.

$$r = clip(\rho(1.25\, z_s + 0.35\, z_r + 0.15\, z_h) + 0.45\, P + 0.30\, S - 0.55\, A - 0.70\, L - 0.35\, N - 0.20\, U, -3, 3) \quad (4.9)$$

This composite can favour holding because holding maximizes stability and preserves the baseline. Novelty is a cumulative intervention penalty, not direct evidence of reduced habituation. These choices are transparent and configurable, but sensitivity analysis and human comfort evaluation are needed before attributing benefit to reward maximization.

4.11 Audio rendering and generated-content management

Intensity maps to a common gain factor between 0.45 and 1. Music and ambience mix scale their separate source levels. Brightness controls a low-pass cutoff between

1200 and 18000 Hz. Tempo maps meditation pitch between 0.92 and 1.08, while density maps ambience pitch between 0.98 and 1.02. Fade changes smoothing and transition duration.

Two meditation sources support crossfaded clip replacement. Recordings are not separated into stems, so density and tempo are playback proxies, not independent compositional controls. Unity does not re-orchestrate the piece after each action.

A generated clip must be decoded and installed before session start. Background requests prepare replacements; cache records retain prompt and model identity. Because replacement changes the composition, affected outcomes are suspended or cancelled during crossfades.

4.12 Persistence and telemetry

Snapshots are isolated by participant, environment, source domain and configuration, with schema validation and atomic rejection of corruption. Simulated and real-window models cannot share weights. Explicit contracts and defensive validation support inspection, but do not constitute security certification.

JSONL records retain the condition, context, proposed and executed actions, predictions, uncertainty, validity, reward terms and update status. Audio and relay session identifiers differ; joins require UTC windows until shared identity binding is verified. Deployment security requires its own acceptance review.

5 IMPLEMENTATION INTEGRATION AND TESTING

The implementation translates the architecture into Unity C# modules and supporting Python utilities. Verification concentrates on numerical correctness, valid learning events and the separation between a decision and its audible execution.

5.1 Development environment and technology stack

Unity owns the runtime scene, audio sources, filters and control lifecycle. C# implements profile processing, the numerical LinUCB model, safety, persistence and telemetry. Python supports supplementary Gymnasium simulations, Stable-Baselines3 experiments and descriptive analysis [11], [12]. The current online model does not require Python inference during playback.

The local generation backend uses FastAPI, Uvicorn and the Google generative-AI client. HTTP requests generate clips and retrieve health information. WebSocket interfaces exist for integration and optional streaming capability, but an available streaming endpoint is not equivalent to verified provider access or continuous music delivery.

Table 5.1 Technology responsibilities

Technology	Component responsibility	Boundary
Unity and C#	Local control, audio rendering, safety and model updates	Runtime on the deployment device
AudioSource and audio filters	Mixing, pitch, brightness and crossfade	Playback rather than composition
Python and Gymnasium	Synthetic environment and experiment orchestration	Supplementary simulation
Stable-Baselines3	PPO, TD3 and SAC implementations	Neural comparison experiments
FastAPI and Uvicorn	Local music request service	Backend process

Technology	Component responsibility	Boundary
Google Lyria API	Preference-conditioned music content	External provider dependency
JSON and JSONL	Profiles, model snapshots and event evidence	Versioned local data contracts

The technical compilation check used the bundled compiler and Unity API references from Unity 6000.4.1f1. This check did not launch, upgrade or validate the full shared project in that editor. The project's declared editor and package configuration remain the authority for an actual build.

5.2 Profile preparation and startup sequence

PreferenceManager loads and normalizes the configured audio preference record. ProfileEngine creates a normalized AudioProfile. The initial content strategy is selected deterministically from preference priors so that identical inputs produce the same starting control configuration across the three study conditions.

PrototypeBootstrap initializes the agent and mixer but holds playback. LyriaClipGenerationService prepares a prompt, submits a request, checks the response, writes audio and metadata, decodes the clip and installs it. Session playback begins only when the personalized generated layer is ready. Generation failure leaves the session stopped with an observable error.

Fixed scene ambience is assigned independently. For a headset, backend addresses must point to the reachable host rather than headset loopback. Successful backend health alone does not confirm clip generation, decode success or audible playback; these are distinct acceptance stages.

5.3 Numerical model and policy implementation

AudioLinUcbModel contains one ridge model per catalogue arm. Selection evaluates only eligible arms, solves the covariance systems with Cholesky decomposition and

returns the selected arm, expected reward, uncertainty and score. Deterministic tie handling makes numerical regression tests repeatable.

Observe validates the reward and feature vector before changing model state. Matrix and vector updates are prepared so an invalid observation does not partially modify an arm. Snapshot restoration checks identity, schema, dimensions and numerical validity before accepting the model.

AudioOnlinePolicy exposes the three complete controller behaviours through a common selection interface. The static policy selects hold. The rule policy selects the eligible catalogue action closest to its conservative target. LinUCB selects the highest eligible score. This separation permits baseline evaluation without inadvertently updating the learned model.

5.4 Agent lifecycle and outcome state machine

AudioRLAgent owns baseline collection, decision timing, pending attribution, reward calculation and model updates. It is evaluated by the bootstrap rather than running a competing independent decision loop. This avoids duplicate evaluations from two owners on the same Unity frame.

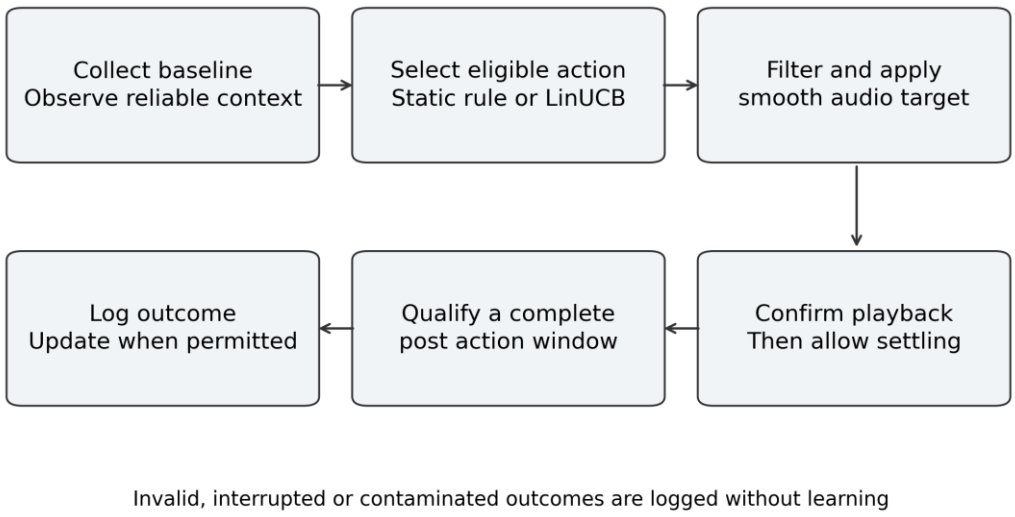


Figure 5.1 Decision and delayed-learning state machine

ObservePlayback receives the mixer's current applied parameters. A pending decision cannot become outcome-eligible until the output reaches the target and settles. AudioOutcomeGate checks temporal and reliability conditions, while AudioSessionBaseline rejects overlapping baseline windows and supplies normalization scales.

The agent reads session lifecycle permission. Acclimatization permits baseline collection without adaptive decisions; the adaptive phase permits qualified decisions; stabilization holds. Pause, termination and network interruption cancel pending outcomes. Resumption requires a fresh full observation window rather than reusing measurements collected during interruption.

5.5 Physiological integration

ComponentBStressSignalReceiver adapts accepted bridge packets to the audio SignalPacket. Nullable HRV availability is preserved. A missing RMSSD is different from a supplied zero and cannot contribute a fabricated improvement term. Stress labels and probabilities remain available upstream, while the audio controller consumes a normalized stress value and its confidence.

The receiver is not an independent validator of sensor accuracy. Its production-domain label describes the input route and window format. The operator must verify that the route is actually driven by the intended wearable and participant, rather than a synthetic source that happens to use the same packet shape.

Clock coherence is important because post-action eligibility is expressed in UTC window time while playback timing uses a monotonic runtime clock. Deployment testing must measure host and device synchronization, window boundaries, routing and network delay. These are necessary conditions for valid attribution, not optional presentation details.

5.6 Playback application and diagnostics

AudioMixerController maintains target and applied parameters and smooths their difference each frame. Meditation and ambience use separate AudioSource objects. A

second meditation source supports crossfading, while optional low-pass and reverb filters shape the meditation playback.

The dashboard displays input signals, controller condition, proposed action, final safe action, latest valid reward, safety state, learning status, update count and output traces. Output traces help confirm signal presence and relative levels. They do not measure listening comfort, acoustic sound pressure or physiological relaxation.

Dashboard mode buttons support demonstrations. Switching condition cancels a pending outcome, returns toward the personalized baseline and labels a new segment. A research session should not use casual mode switching as a substitute for randomized or counterbalanced condition assignment.

5.7 Persistence replay and evidence protection

AudioBanditPersistence hashes the participant, environment and source-domain identity, with a separate configuration key. Snapshots include feature and action schema information. Writes use a temporary file and atomic replacement where supported. Corrupt, oversized or incompatible files are rejected rather than trusted.

AudioRLTransitionLogger writes version-2 JSONL decision and outcome events. AudioRLReplayBuffer copies records on insertion and retrieval so later callers cannot mutate stored evidence. Invalid outcomes retain their reason and update status without contributing a fabricated reward to the learning model.

Hashing an identifier does not anonymize physiological data. Logs and generated-content metadata still need access controls, retention rules and a deletion procedure. Participant models should be treated as research data, not harmless application cache files.

5.8 Verification strategy

The verification suite uses independent test expectations for the numerical model and explicit Unity test doubles for standalone agent execution. It checks known regression values, action masking, corrupted snapshot rejection and model isolation. Runtime tests cover delayed attribution, playback convergence, failure handling and non-learning conditions.

The Python tests inspect result aggregation rather than controller mathematics. They verify participant-level averaging, deduplication, exclusion of simulated data by default, invalid-only sessions and the absence of a manufactured standard deviation for a single participant.

Table 5.2 Verification groups and evidence

Group	Checks performed	Evidence boundary
Numerical selection	Ridge estimate, uncertainty, ties and masks	Mathematical implementation
Model integrity	Invalid updates, restore validation and isolation	Local snapshot behaviour
Feature and action contract	Forty features, thirteen arms, optional fields	Schema consistency
Safety	Hold, finite values and confidence restrictions	Software bounds
Delayed learning	Settle, non-overlap, uniqueness and timeout	Standalone agent lifecycle
Experimental modes	Static, rule, learning and frozen evaluation	Update permissions
Evaluation analysis	Participant weighting, missingness and source separation	Descriptive results pipeline

5.9 Deployment and acceptance procedure

Before a participant session, the operator verifies the scene, fixed ambience, assigned condition, profile identity, generated meditation clip and live signal route. The model domain must be physiological, with simulated learning disabled for the real-user protocol. Model reset or persistence follows the declared study design.

The actual Unity editor verification, Play Mode, headset build and end-to-end wearable run are separate acceptance gates. The operator tests start, pause, resume, emergency mute, network loss, generation failure and model reload on the target device. A session proceeds only after the deployment checks pass and the participant remains comfortable.

6 RESULTS AND VALIDATION

This chapter reports completed technical evidence and a supplementary synthetic algorithm comparison. Participant outcomes are not yet available. The evidence is sufficient to describe implementation correctness within the tested scope, but not to conclude that the controller improves physical relaxation.

6.1 Completed online-agent verification

The standalone verification run on 11 October 2026 passed 22 C# tests. A separate Python suite passed five evaluation tests. All audio runtime and editor sources compiled against the Unity API references used by the verification harness. No compilation errors occurred; warnings concerned serialized fields normally assigned through the Inspector.

Table 6.1 Completed technical verification results

Verification activity	Observed result	Interpretation
Audio source compilation	Passed	Source compatible with checked API references
Standalone C# tests	22 passed	Numerical, safety and agent checks passed
Python evaluation tests	5 passed	Aggregation and filtering checks passed
Synthetic contextual sanity check	17 of first 20 and 20 of last 20 preferred choices	Small deterministic software-learning example
Unity Play Mode and headset validation	Not reported	Requires deployment execution
Real-user comparative study	Not yet conducted	No human efficacy result available

The contextual sanity check demonstrates that the update mechanism can change choices toward a programmed reward relation. It is not an audio experiment and does not establish generalization or relaxation. The 27 passed tests are test cases, not 27 participants or 27 independent physiological outcomes.

6.2 Numerical and integrity results

A known ridge-regression case produced the expected mean of 0.5 and uncertainty equal to the square root of 0.5. Invalid observations left the model unchanged. Eligible-action masking and deterministic ties behaved as specified. Corrupt-snapshot restoration was rejected without a partial state change.

Model isolation tests separated participant, environment, configuration and source domain. The 40-feature encoding and all 13 catalogue actions passed their contract checks. Hold did not repeat a preceding action. Missing RMSSD did not create a physiological reward component.

These results support the integrity of the implemented decision mechanism. They do not validate the linear reward assumption, the exploration coefficient or the appropriateness of action sizes for every listener.

6.3 Delayed learning and analysis results

Runtime tests confirmed that an action updated the model once only after target playback and a qualified post-window. Duplicate, overlapping, stale, future, low-quality and expired observations did not train the model. Pause, emergency and contaminated outcomes were cancelled. Frozen evaluation calculated rewards without changing weights.

The analysis tests confirmed that repeated sessions are averaged within participant rather than allowing prolific participants to dominate a group mean. Invalid-only sessions were retained without invented reward means. Simulated and physiological comparisons remained separate, and one-participant standard deviations remained unavailable.

These are controlled software tests with explicit test doubles. Their success does not demonstrate actual device timing, network reliability, perceptual smoothness or

wearable correctness. Those properties belong to deployment and participant validation.

6.4 Supplementary preference-only simulation protocol

A separate Python experiment compared a rule baseline, a linear contextual-bandit baseline, PPO, TD3 and SAC using synthetic users. Per seed, the configured split comprised 64 training users, 16 validation users and 24 held-out test users. Seeds were 11, 23 and 37. Each test user received five 120-step episodes.

Neural algorithms used a configured 120000-step budget per run. Rule and bandit runs used 150 episodes, or 18000 steps when complete. Total interaction budgets therefore differ across methods. The neural benchmark evaluated saved final policies, not uniformly selected best checkpoints.

The simulator supplied 34 observations and seven continuous actions, combining a conservative rule term with model adjustments before filtering. Observations included hidden ideal targets and latent responsiveness. This is an optimistic controlled testbed, not a faithful representation of information available from participants.

The supplementary linear bandit is distinct from the deployed 40-feature, 13-action LinUCB controller. Simulator feedback, actions and rewards differ. These results cannot validate the deployed policy or share a comparative table with participant outcomes.

6.5 Supplementary algorithm results

The aggregate CSV reports means and population standard deviations across three runs, using a divisor of three. These are not participant uncertainty intervals. Seeds regenerate users, so variability includes both learning randomness and synthetic population differences.

Table 6.2 Supplementary simulation reward and preference results

Method	Episode reward mean	Run SD	Preference score
Rule baseline	248.317	0.648	0.907970
Linear bandit baseline	221.386	2.956	0.927375
PPO	242.999	8.562	0.924582
TD3	189.330	2.641	0.760426
SAC	215.899	1.368	0.966014

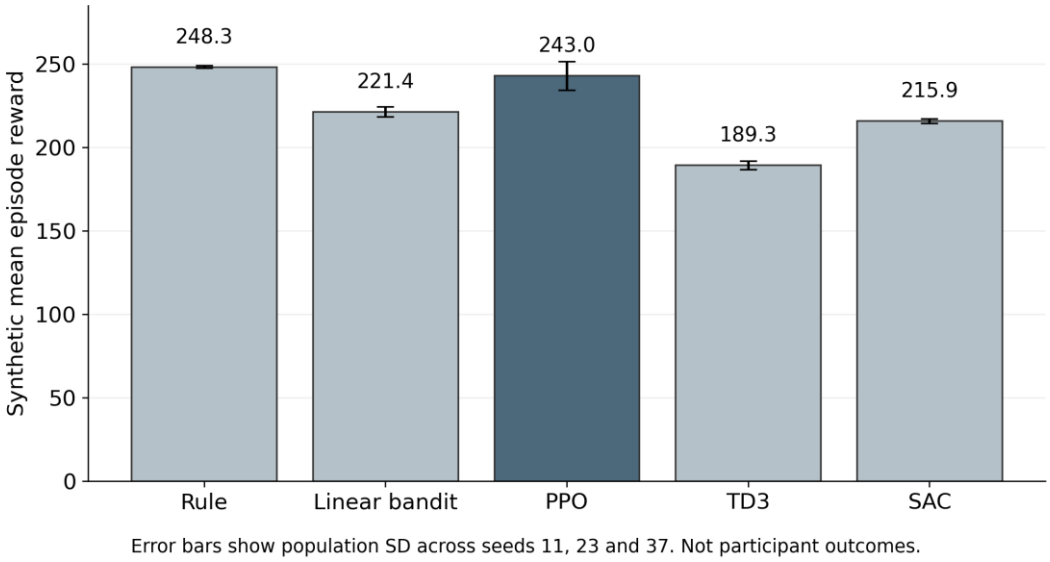


Figure 6.1 Supplementary synthetic episode reward across three runs

The rule baseline achieved the highest composite reward. PPO led the neural methods on reward; SAC led on preference score. Different metrics therefore rank controllers differently. No universal algorithm superiority or human relaxation effect follows.

Table 6.3 Supplementary simulation control metrics

Method	Proposal threshold rate	Smoothness score	Intervention frequency
Rule baseline	0.000000	0.999390	0.012222
Linear bandit baseline	0.000278	0.998743	0.043009
PPO	0.000000	0.999251	0.015671
TD3	0.483519	0.999004	0.020995
SAC	0.067060	0.998214	0.028264

The stored field named safety_violation_rate counts proposed actions exceeding 1.5 times the simulator action limit before filtering. It is labelled proposal threshold rate here to avoid implying that a listener experienced an unsafe output. Smoothness equals one minus the simulator abrupt-change penalty, while intervention frequency counts steps whose mean safe-action magnitude exceeds 0.01.

The simulator already starts at the hidden preference target and contains a relaxation-directed drift. This can favour conservative behaviour and inflate apparent preference alignment. The observations and reward also share preference-derived quantities. These characteristics limit interpretation of the high scores and motivate reporting component outcomes rather than reward alone.

6.6 Participant and integration validation status

Real-user testing is planned after the present submission. There are no completed participant counts, satisfaction percentages, physiological effect sizes, confidence intervals or significance tests to report. The questionnaire and counterbalanced protocol are specified so that forthcoming evidence can be added without changing the research question after observing the results.

The implemented integration includes profile preparation, clip startup gating, accepted-window adaptation, safety, playback targets, logging and persistence. The available verification does not establish that this chain has passed an end-to-end real-

wearable headset session. Deployment logs and participant observations are required to close that evidence gap.

7 DISCUSSION

The discussion interprets the evidence in relation to the research questions and identifies where implementation correctness stops short of a research conclusion about listeners.

7.1 Interpretation of technical findings

The test results show that learning updates can be restricted to a valid delayed outcome and that invalid events need not contaminate personalized weights. This is essential because the numerical learner itself cannot recognize whether a reward came from an overlapping window, a paused session or a different participant.

Playback-aware attribution is particularly important. A target parameter assignment is not an audible intervention until the mixer applies it. Beginning the outcome window after actual convergence and settling is more defensible than rewarding an action from the next arbitrary packet. It also reduces the number of usable updates, which must be measured rather than hidden.

7.2 Interpretation of algorithm choice

The choice of LinUCB is supported by the controller's limited action catalogue, sparse outcomes and need for inspection. It is not supported by a completed human comparison showing LinUCB to be best. Separate action models make the selection rule transparent, but they can be data hungry when each participant contributes only a few valid observations.

Personalization has two levels. The preference pipeline supplies immediate personalization before any learning. Online weights then distinguish the participant's observed responses to safe actions. A participant-specific filename alone does not demonstrate useful personalization; the study must show sufficient valid updates and improved choices or experience.

The linear assumption can miss nonlinear interactions between preference, stress and sound. The present implementation also lacks explicit forgetting, so a model can

accumulate outdated evidence across sessions. Future work can examine confidence-bounded resets, recency weighting or shared priors, but these require controlled evaluation rather than automatic addition.

7.3 Comparison with literature and baselines

The system complements VR mindfulness research by making the audio controller and its evidence boundary explicit [1], [2]. The music literature motivates sensitivity to stimulus and person rather than a universal calming configuration [3]. The bandit literature supplies a decision method, but does not guarantee its assumptions in delayed physiological control [5].

The supplementary benchmark reinforces the value of strong non-learning baselines. A high neural reward cannot be assumed from model complexity, and high preference alignment can coexist with lower composite reward. The real-user study must therefore retain both Static Personalized and Rule-Based Adaptive rather than only comparing learned algorithms.

7.4 Research contribution

The research contribution is a bounded online personalization mechanism with explicit separation of physiological confidence, model uncertainty and operational permission. It also specifies how actual playback completion, qualified pre/post windows and contamination cancellation determine when an action is learnable.

A further contribution is the evaluation boundary. Reward, profile alignment, filter intervention and perceived relaxation are distinct quantities. Reporting them separately reduces the risk of presenting engineering proxies as human benefit. The participant-level aggregation pipeline supports this separation without pretending that repeated windows increase the participant sample size.

7.5 Product contribution

The product artefact connects preference-aligned generated music with fixed scene ambience and local adaptive control. It provides a strict generated-content startup gate, confidence-aware holds, crossfade management, observable controller modes

and per-participant persistence. These capabilities make a supervised prototype demonstrable and inspectable.

The operational design allows generation services to fail without silently relabelling unrelated audio as personalized generated music. Local control remains separate from provider content generation. This boundary supports future content services while retaining the same safety and learning interface.

7.6 Limitations and threats to validity

Construct validity is limited by the reward. Preference match is a distance to the initialized baseline, stability favours small actions, and the novelty penalty is a cumulative change counter. None directly measures meditation quality or habituation. HR and RMSSD direction are imperfect proxies and may reflect breathing, movement or spontaneous recovery.

Internal validity is threatened by composition changes, visual adaptation, session order, baseline drift, regression to the mean and expectation effects. An eligible window is temporally compatible with an action but does not prove causation. Holding visuals and content fixed and counterbalancing session order reduce, rather than eliminate, these threats.

Statistical validity is limited by few participants and sparse qualified outcomes. Forty features for each of 13 disjoint arms can be ambitious for a short session. Conservative eligibility can further reduce action diversity. The pilot should report per-arm counts, update coverage and missingness before making personalization claims.

External validity is limited to one scene, the tested content and the recruited population. Generated music may vary across requests, cultural expectations differ and headset acoustics affect comfort. A result in healthy university volunteers would not establish efficacy for clinical populations or other environments.

Engineering limitations include incomplete target-device acceptance evidence, local rather than explicitly shared session identifiers, provisional thresholds and playback

proxies for musical controls. Authentication, encrypted transport, access control and deletion must be verified before use beyond supervised research.

7.7 Implications for the forthcoming study

The immediate study should prioritize feasibility and comfort before optimizing reward. Its first questions are whether the correct user receives the correct content, valid windows arrive, actions become audible smoothly and enough clean outcomes exist to permit learning. A session with many cancellations is an important result, not a data-cleaning inconvenience.

The controller can remain a useful personalized audio product even if the learned condition does not outperform a rule baseline. Such a finding would constrain the value of online learning under the available signal rate and reward, rather than invalidate preference-aligned generation or safe mixing. Negative and inconclusive results should therefore be retained.

8 COMMERCIALISATION AND SUSTAINABILITY

Commercial potential is considered as a future pathway rather than evidence of market readiness. The current artefact is a supervised research prototype with unvalidated participant benefit and deployment requirements.

8.1 Value proposition and target users

The proposed value is an immersive meditation experience that respects personal sound preferences and changes conservatively with reliable context. Potential users include university wellbeing programmes, research laboratories and supervised wellness facilities. These are candidate application settings, not confirmed customers.

The strongest immediate value is research infrastructure: explainable control, auditable observations and repeatable comparisons. A consumer value proposition would require demonstrated comfort, usability, willingness to use and acceptable operating costs. Clinical positioning would require a substantially different validation and regulatory pathway.

8.2 Market and competitive position

The architecture differs conceptually from a fixed audio library by combining scene-aware content preparation with responsive playback. It differs from an unconstrained generative music demonstration by including a qualified feedback loop and deterministic safety. No market-share estimate, competitor performance ranking or customer demand figure is claimed.

A competitive evaluation should compare setup burden, content consistency, offline behaviour, personalization quality, privacy and cost. The relevant alternative is not only another AI system; it is also a well-chosen static recording or a simple rule-based mixer. The participant study helps determine whether the added complexity is justified.

8.3 Operating and sustainability model

Potential operating models include institutional licensing, supervised research deployments or optional content-generation services. Pricing is not proposed without usage measurements and provider terms. The cost model should include generated clips, retries, storage, network service, headset support and maintenance rather than API calls alone.

Caching content and reusing a participant-approved clip can reduce unnecessary requests. Local LinUCB updates avoid a separate neural inference service during playback. These are resource-conscious design choices, but measured energy savings or reduced cloud expenditure have not been established.

8.4 Scalability intellectual property and compliance

The component can support additional environments through separate ambience assignments and scene-aware prompt construction. Generalization of control limits still requires scene and device evaluation. Multiple participants require verified isolation, authenticated routing, concurrency control and secure model storage.

Intellectual property considerations include ownership of source code, university policy, third-party package licences and provider terms for generated music. Generated output should not be assumed to have unrestricted commercial rights. Participant physiology and personalized models require explicit consent, controlled retention and an appropriate data-protection review.

8.5 Contribution to sustainable development

The research is relevant to Sustainable Development Goal 3 through its intended support for wellbeing research [16]. This is an alignment of purpose, not a demonstrated public-health impact. Accessibility, safe use and affordability determine whether the eventual product provides meaningful benefit.

Responsible sustainability also includes minimizing unnecessary data collection, avoiding misleading health claims and providing a useful non-adaptive fallback. A smaller, explainable controller can be preferable to computational complexity when its performance and usability are adequate.

9 CONCLUSION AND FUTURE DIRECTIONS

The thesis establishes an implemented online audio-control architecture and a transparent boundary between verified software behaviour and the human evidence still required.

9.1 Conclusion

A confidence-aware personalized adaptive audio controller has been implemented for the Japanese Temple Pond Garden meditation setting. The system prepares generated instrumental music from user preferences and scene context, retains fixed environment ambience and uses a local disjoint LinUCB contextual bandit to select safe micro-adjustments.

The design combines a versioned 40-element context, 13 bounded actions, deterministic safety, actual-playback feedback and delayed physiological qualification. Participant-specific snapshots and machine-readable evidence support ongoing learning and inspection. Twenty-seven automated tests and the source compilation check support technical correctness within their stated scope.

Human benefit and comparative superiority remain unestablished. The forthcoming study must assess comfort, personalization, usable feedback rate and session outcomes. The contribution is a technically verified research artefact and specified evaluation method, not a clinically validated intervention.

9.2 Achievement of objectives

Table 9.1 Objective achievement and remaining evidence

Objective	Achievement	Remaining evidence
Preference-aligned initialization	Profile and deterministic strategy implemented	Participant preference confirmation
Scene-aware generated content	Prompt, service, cache and startup gate implemented	Target-deployment generation acceptance

Objective	Achievement	Remaining evidence
Online personalized selection	LinUCB and persistence implemented and tested	Real-user learning exposure and benefit
Confidence-aware bounded control	Safety and masking checks passed	Listening comfort and acoustic review
Valid delayed learning	Playback and window tests passed	Actual-device timing and live delivery
Comparative evaluation	Three modes and descriptive analysis implemented	Counterbalanced participant sessions
Reproducible evidence	Schemas, tests, logs and analysis tools available	Final study dataset and inferential report

9.3 Individual research and product contributions

The individual work contributes the audio-control model, action and feature design, reliability-aware reward, outcome attribution, local persistence, generated-content orchestration and research telemetry. Its principal architectural contribution is that a learned proposal becomes a training observation only after safe execution and valid delayed measurement.

The Unity prototype supports supervised comparison of static, rule-based and learned adaptation. Numerical, synthetic and prospective evidence remain distinct, separating application functionality from demonstrated participant benefit.

9.4 Immediate and longer-term research

The immediate priority is target-device acceptance and the approved participant pilot. This includes identity routing, genuine wearable input, timing, comfortable playback, stop behaviour and log completeness. Reward weights, action sizes and confidence thresholds should be frozen before the comparative study or recorded as a separate pilot-calibration phase.

Longer-term research should examine repeated-session personalization, smaller or better-conditioned feature sets, shared priors with participant isolation, nonstationarity and confidence calibration. Richer DSP or stem-based music can provide more faithful tempo and density control. Deep continuous-control methods can be reconsidered when representative sequential data justify their complexity.

Future streaming generation requires independent latency, reliability, cost and quality evaluation without weakening safety or attribution. Stronger relaxation or treatment claims require longitudinal and appropriately approved clinical research.

REFERENCES

[1] M. Modrego-Alarcon et al., "Efficacy of a mindfulness-based programme with and without virtual reality support to reduce stress in university students: A randomized controlled trial," Behaviour Research and Therapy, vol. 142, Art. no. 103866, 2021. doi: 10.1016/j.brat.2021.103866. Available: https://pubmed.ncbi.nlm.nih.gov/33957506/

[2] M. V. Navarro-Haro et al., "Meditation experts try Virtual Reality Mindfulness: A pilot study evaluation of the feasibility and acceptability of Virtual Reality to facilitate mindfulness practice in people attending a Mindfulness conference," PLOS ONE, vol. 12, no. 11, Art. no. e0187777, 2017. doi: 10.1371/journal.pone.0187777. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC5699841/

[3] M. V. Thoma, R. La Marca, R. Bronnimann, L. Finkel, U. Ehlert and U. M. Nater, "The effect of music on the human stress response," PLOS ONE, vol. 8, no. 8, Art. no. e70156, 2013. doi: 10.1371/journal.pone.0070156. Available: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0070156

[4] S. Laborde, E. Mosley and J. F. Thayer, "Heart rate variability and cardiac vagal tone in psychophysiological research: Recommendations for experiment planning, data analysis, and data reporting," Frontiers in Psychology, vol. 8, Art. no. 213, 2017. doi: 10.3389/fpsyg.2017.00213. Available: https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2017.00213/full

[5] L. Li, W. Chu, J. Langford and R. E. Schapire, "A contextual-bandit approach to personalized news article recommendation," in Proceedings of the 19th International Conference on World Wide Web, 2010, pp. 661-670. doi: 10.1145/1772690.1772758. Available: https://arxiv.org/abs/1003.0146

[6] J. Garcia and F. Fernandez, "A comprehensive survey on safe reinforcement learning," Journal of Machine Learning Research, vol. 16, pp. 1437-1480, 2015. Available: https://jmlr.org/papers/v16/garcia15a.html

[7] K. Peffers, T. Tuunanen, M. A. Rothenberger and S. Chatterjee, "A design science research methodology for information systems research," Journal of Management Information Systems, vol. 24, no. 3, pp. 45-77, 2007. doi: 10.2753/MIS0742-1222240302. Available: https://doi.org/10.2753/MIS0742-1222240302

[8] J. Schulman, F. Wolski, P. Dhariwal, A. Radford and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017. Available: https://arxiv.org/abs/1707.06347

[9] S. Fujimoto, H. van Hoof and D. Meger, "Addressing function approximation error in actor-critic methods," in Proceedings of the 35th International Conference on Machine Learning, PMLR, vol. 80, 2018, pp. 1587-1596. Available: https://proceedings.mlr.press/v80/fujimoto18a.html

[10] T. Haarnoja, A. Zhou, P. Abbeel and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in Proceedings of the 35th International Conference on Machine Learning, PMLR, vol. 80, 2018, pp. 1861-1870. Available: https://proceedings.mlr.press/v80/haarnoja18b.html

[11] A. Raffin et al., "Stable-Baselines3: Reliable reinforcement learning implementations," Journal of Machine Learning Research, vol. 22, no. 268, pp. 1-8, 2021. Available: https://jmlr.org/papers/v22/20-1364.html

[12] M. Towers et al., "Gymnasium: A standard interface for reinforcement learning environments," arXiv:2407.17032, 2024. Available: https://arxiv.org/abs/2407.17032

[13] Google, "Generate music with Lyria 3.5," Gemini API documentation. Accessed Oct. 11, 2026. Available: https://ai.google.dev/gemini-api/docs/music-generation

[14] Unity Technologies, "AudioSource pitch," Unity 6 scripting reference. Accessed Oct. 11, 2026. Available: https://docs.unity3d.com/6000.0/Documentation/ScriptReference/AudioSource-pitch.html

[15] World Medical Association, "WMA Declaration of Helsinki: Ethical principles for medical research involving human participants," 2024 revision. Accessed Oct. 11, 2026. Available: https://www.wma.net/policies-post/wma-declaration-of-helsinki/

[16] United Nations, "Goal 3 Ensure healthy lives and promote well-being for all at all ages," Sustainable Development Goals. Accessed Oct. 11, 2026. Available: https://sdgs.un.org/goals/goal3

[17] L. Li, W. Chu, J. Langford and X. Wang, "Unbiased offline evaluation of contextual-bandit-based news article recommendation algorithms," arXiv:1003.5956, 2010. Available: https://arxiv.org/abs/1003.5956

GLOSSARY

Audio profile. A normalized representation of persistent sound preferences and the initialized control configuration.

Contextual bandit. A learner that selects an action from observed context and updates from feedback for the selected action, without modelling a full long-horizon transition process.

Disjoint model. A separate regression model for each catalogue action rather than one shared coefficient vector across all actions.

Eligible action. An action permitted under the current deterministic constraints before model selection.

Delayed outcome. A qualified observation collected after the action is audibly applied and the settling interval has elapsed.

Physiological confidence. The predictive certainty attached to the supplied stress estimate, distinct from measurement quality or model uncertainty.

Model uncertainty. The LinUCB exploration quantity associated with limited observations in the current context.

Preference proximity. A mathematical distance-based proxy for profile consistency, not a direct rating of enjoyment or satisfaction.

Source domain. The separation between simulated observations and accepted physiological-window observations.

Whole-controller condition. A complete static, rule-based or learned decision method evaluated through the same playback and safety path.

APPENDICES

Appendix A Reproducible controller specification

The active context schema is audio-context-v1-40 and the action schema is audio-micro-actions-v1. Covariance matrices have 1600 values per arm and reward vectors have 40 values per arm. Thirteen arms contain 21320 numeric coefficients before metadata, approximately 166.6 KiB as raw 64-bit values. JSON storage is larger. This is a size calculation, not a measured serialized-file benchmark.

Table A.1 Timing and learning configuration

Parameter	Default	Qualification
Exploration alpha	0.25	Used only when exploration enabled
Ridge lambda	1.0	Positive prior covariance scale
Baseline window count	2	Independent accepted windows required
Production decision minimum	75 seconds	Pending outcome can extend actual interval
Settling duration	5 seconds	Starts after playback reaches target
Outcome timeout	240 seconds	Cancellation without weight update
Production input age limit	90 seconds	Freshness check
Accepted window duration	30 to 120 seconds	Input validity range
Learning confidence minimum	0.45	Both relevant windows qualified
Learning quality minimum	0.60	Both relevant windows qualified

Parameter	Default	Qualification
Simulation decision interval	5 seconds	Separate demonstration domain
Replay capacity	2048 outcomes	Bounded copied records

The context indices 1 to 13 are, in order: intensity, density, brightness, tempo, music mix, ambient mix, rhythm amount, nature level, reverb amount, novelty tolerance, dissonance allowance, responsiveness prior and confidence-sensitivity prior. The applied-control order is intensity, density, brightness, tempo, fade, music mix and ambient mix.

The default episode horizon is 120 decision opportunities. This normalizes a progress feature; it is not a 120-second meditation duration. Learning permission, simulation permission and persistence are independent settings. Real-user operation must explicitly verify their deployed values.

Appendix B Physiological and generation contracts

The upstream physiological record contains timestamp, heartRate, rmssd, sdnn, stress, signalQuality, windowStart and windowEnd. The stress object can include a band label, low and high classes, confidence, adjacent-class information, probabilities and continuous_score. Component E consumes the accepted normalized representation rather than inferring a categorical label from free text.

Table B.1 Input fields and audio usage

Input	Meaning	Use in Component E
timestamp and window boundaries	Measurement timing	Freshness and non-overlap
Stress estimate	Normalized current stress	Context, rules and reward
Confidence	Stress-estimate certainty	Eligibility, safety and reliability
signalQuality	Measurement usability	Eligibility, safety and reliability

Input	Meaning	Use in Component E
heartRate	Optional beats per minute	Flagged context and optional reward
rmssd	Optional milliseconds	Flagged context and optional reward
sdnn	Optional milliseconds	Preserved signal information

The generation request contains a prompt, configured model, request identifier and instrumental-only intent. A representative prompt requests original instrumental meditation for a quiet Japanese temple pond garden, the preferred instruments, slow pacing, gentle consonant harmony and restrained brightness. It avoids claiming that music can medically treat the participant.

The local backend provides GET /health, POST /generate-clip, GET /realtime-capability and WS /live-music. Endpoint existence does not prove model access. Successful personalized playback requires a successful generation response, valid audio bytes, decoding, clip assignment and the explicit session start gate.

Appendix C Participant instruments and operator record

The following items are prospective research instruments. They are designed for a feasibility pilot and are not claimed to be validated psychometric scales. The same wording, response anchors and administration order should be used for every controller condition.

Before the session, ask: "How stressed do you feel right now?" Use 0 for not stressed at all and 10 for extremely stressed. Record current comfort and willingness to proceed. Collect sound preferences before the first condition, with changes documented rather than silently remapped between conditions.

After the session, repeat the current-stress item and ask the participant to rate the following from 1, strongly disagree, to 5, strongly agree. Provide a separate not applicable option. Do not replace a missing answer with the midpoint.

- The meditation audio matched my usual sound preferences.
- The audio suited the Japanese Temple Pond Garden setting.
- Audio changes felt smooth and natural.
- Audio changes distracted me from meditation.
- I felt physically relaxed during the session.
- The balance between meditation music and ambience was comfortable.
- I would choose to use this audio experience again.

Ask an open question: "Which sound or transition would you change, and why?" Record discomfort, unexpected sound, interruption, request to stop and the participant's explanation in their own words. A positive questionnaire response is not substituted for physiological evidence, and the physiological reward is not substituted for a questionnaire response.

The operator record includes pseudonymous ID, assigned condition, order, UTC start and end, clip hash, profile revision, model-reset policy, initial update count, final update count, source domain, interruptions, valid outcomes and invalidity reasons. Any manually altered audio or environment setting is recorded as a potential confound.

Appendix D Reproduction and analysis procedure

The source root is R26-DS-006. The audio runtime resides under unity/Assets/AdaptiveAudioVR. The local music service resides under services/lyria_backend. The separate simulation source and configuration reside under tools/rl_training, with experiment outputs retained as their own evidence set.

Run the C# verification from the repository root using the following command. It rebuilds the checked references and runs the standalone harness. It is not a headset build or a substitute for Unity engine verification.

```
powershell -NoProfile -ExecutionPolicy Bypass -File tools/test-audio-online.ps1
```

Run the descriptive evaluation tests using a Python environment with the repository available as the working directory.

```
python -m unittest discover -s tests -p test_audio_online_evaluation.py -v
```

After collection, point the evaluation utility at the actual RLTransitions directory. Simulation records are excluded by default. The output includes sessions.csv, participants.csv, comparison.csv, comparison.json and report.md. The utility does not conduct the prospective inferential tests described in Chapter 3.

```
python tools/evaluate-audio-online.py --logs "C:\Study\RLTransitions" --output
results/audio_online
```

For a Unity verification run, use AdaptiveAudioVR then Verify Audio RL Agent. On AdaptiveAudioIntegration, select the Audio Study Condition in the scene installer. Research runs use one assigned condition per session rather than dashboard switching. Generated content, stable user identity and live-window routing are confirmed before exposure.

Retain a manifest containing the code revision, package versions, configuration hash, content hashes, test outputs, assigned order and model snapshots. Store participant consent and identifying data separately. An analyst should be able to trace a reported outcome to one decision, one executed action and one qualified post-window.

Appendix E Academic integrity and AI use disclosure

Generative AI assistance was used to help organize the thesis, draft and revise explanatory text, inspect implementation structure and prepare document-generation and verification workflows. The author is responsible for checking every claim, reference, mathematical expression, result and description before submission, in accordance with the university's applicable policy.

Representative assisted tasks included explaining confidence-aware action filtering, specifying delayed-outcome attribution, comparing learning alternatives and formatting technical tables. The document distinguishes verified automated checks, supplementary synthetic experiments and prospective human evaluation. AI assistance does not supply missing participant data, ethics approval or independent clinical validation.

Google Lyria is also part of the research artefact and generates meditation audio from prompts. This functional use is distinct from assistance with writing or coding. Source citations, configuration records and retained experiment outputs provide the

basis for verification. The author should amend this disclosure if the institution requires additional detail about tool versions or specific contributions.